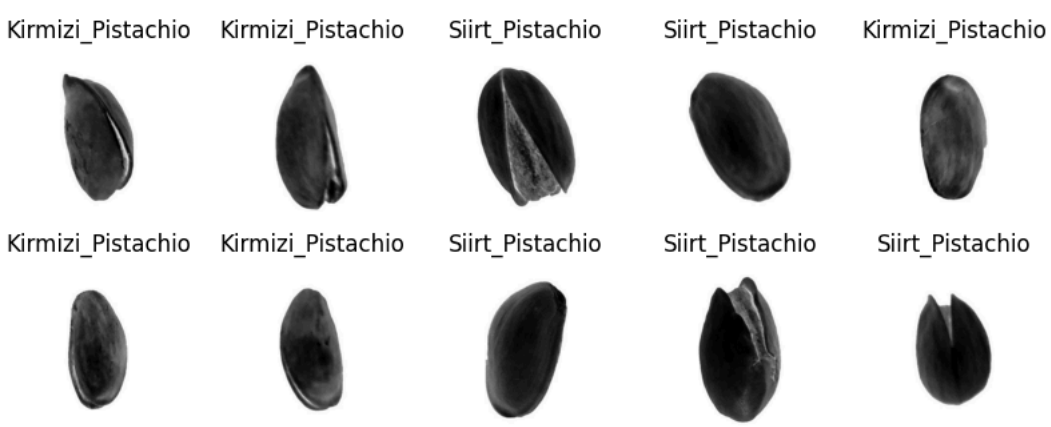
```
import os
import torch
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
from PIL import Image
import matplotlib.pyplot as plt
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
dataset_path = "/content/drive/MyDrive/ANNlab/Pistachio_Image_Dataset/Pistachio_Image_Dataset"

# Define image transformations
transform = transforms.Compose([
    transforms.Resize((256, 256)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.5], std=[0.125])
])



# apply multiclass classifcation
class PistachioDataset(Dataset):
    def __init__(self, root_dir, transform=None):
        self.root_dir = root_dir
        self.transform = transform
        self.images = []
        self.labels = []

         #Load images from "Kirmizi_Pistachio" folder (Class 0)
        kirmizi_folder = os.path.join(root_dir, "Kirmizi_Pistachio")
        for file in os.listdir(kirmizi_folder):
            if file.lower().endswith(('jpg', 'png', 'jpeg')):
                self.images.append(os.path.join(kirmizi_folder, file))
                self.labels.append(0)  # 0 for Kirmizi_Pistachio

        # Load images from "Siirt_Pistachio" folder (Class 1)
        siirt_folder = os.path.join(root_dir, "Siirt_Pistachio")
        for file in os.listdir(siirt_folder):
            if file.lower().endswith(('jpg', 'png', 'jpeg')):
                self.images.append(os.path.join(siirt_folder, file))
                self.labels.append(1)  # 1 for Siirt_Pistachio

    def __len__(self):
        return len(self.images)

    def __getitem__(self, idx):
        # Load image and label
        img_path = self.images[idx]
        label = self.labels[idx]

        # Open image
        image = Image.open(img_path).convert("L")

        # Apply transformations
        if self.transform:
            image = self.transform(image)

        # Convert label to tensor
        label = torch.tensor(label, dtype=torch.long)

        return image, label

# Instantiate dataset
dataset = PistachioDataset(dataset_path, transform=transform)

# Create DataLoader
dataloader = DataLoader(dataset, batch_size=64, shuffle=True)

# Test the DataLoader
for images, labels in dataloader:
    print(f"Batch size: {images.shape}, Labels: {labels}")
    break
```

```
Batch size: torch.Size([64, 1, 256, 256]), Labels: tensor([0, 0, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0,
        0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0,
        0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1])
```

```
image,labels = next(iter(dataloader))
class_name = ['Kirmizi_Pistachio','Siirt_Pistachio']
plt.figure(figsize = (10,10))
for i in range(10):
    plt.subplot(5,5,i+1)
    img = image[i].permute(1,2,0)
    img = img *0.123 + 0.5
    plt.imshow(img.numpy(),cmap = 'Grays')
    plt.title(f"{class_name[labels[i].item()]}")
    plt.axis("off")

plt.show()
```




```
#applying regrssion model
import keras
from keras import layers
from keras.models import Sequential
import numpy as np

# Define the model
model = Sequential()

# Input layer (flattened)
model.add(layers.Input(shape=(65536,)))

# Hidden layer with ReLU activation
model.add(layers.Dense(1024, activation='relu'))

# Output layer with ReLU activation
model.add(layers.Dense(65536, activation='relu'))

# Compile the model
model.compile(optimizer='adam', loss='mean_squared_error')

# Example usage
if __name__ == "__main__":
    # Create a random input tensor with shape [10, 1, 256, 256]
    input_tensor = images

    # Flatten the input tensor to shape [10, 65536]
    input_tensor_flat = input_tensor.reshape(64, 65536)
    n_epochs =100
    for epoch in range(n_epochs):
        # Forward pass through the model
        output_tensor_flat = model.predict(input_tensor_flat)

    # Reshape the output back to [10, 1, 256, 256]
    output_tensor = output_tensor_flat.reshape(64, 1, 256, 256)

    # Print the output shape
    print(output_tensor.shape)
```

```
2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 267ms/step
2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 257ms/step
2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 311ms/step
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2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 249ms/step
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2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 456ms/step
2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 408ms/step
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2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 398ms/step
2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 445ms/step
2/2 ━━━━━━━━━━━━━━━━━━━━ 1s 463ms/step
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2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 257ms/step
```

```
type(output_tensor)
tensor = torch.from_numpy(output_tensor)
print(tensor.shape)
```

```
torch.Size([64, 1, 256, 256])
```

```
import matplotlib.pyplot as plt
plt.figure(figsize = (10,10))
for i in range(64):
  plt.subplot(8,8,i+1)
  img2 = tensor[i].permute(1,2,0)
  img2 = img2 *0.125 + 0.5
  plt.imshow(img2.numpy(),cmap = 'Grays')
  plt.axis("off")

plt.show()
```

Start coding or generate with AI.